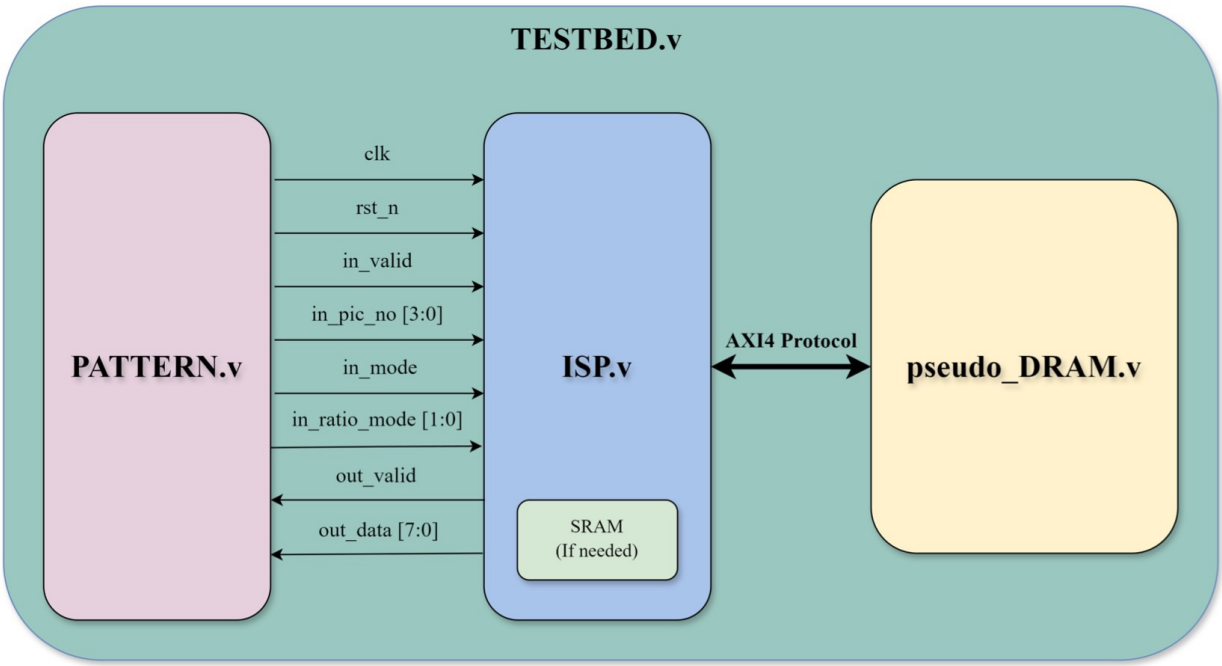


Fig. 2: ISP System for midterm project.

TABLE I: Signal Description between PATTERN.v and ISP.v

Signal	Width	From	Description
clk	1	PATTERN.v	System clock.
rst_n	1	PATTERN.v	Reset all system.
in_valid	1	PATTERN.v	When in_valid=1, means the signal in_mode, in_pic_no and in_ratio_mode signals are valid. Will be high for only 1 cycle.
in_pic_no	4	PATTERN.v	Select which picture from the DRAM. Only valid when in_valid=1
in_mode	2	PATTERN.v	Select algorithm type. 0 for Auto Focus, 1 for Auto Exposure, and 2 for Average of min max value. Only valid when in_valid=1.
in_ratio_mode	1	PATTERN.v	Select the ratio for auto exposure. 0 means 0.25, 1 means 0.5, 2 means 1, 3 means 2. Only valid when in_valid=1 and in_mode=1.
out_valid	1	ISP.v	When out_valid=1, means the our_data is valid. Note that cannot be overlapped with in_valid.
out_data	8	ISP.v	Return the best contrasts when algorithm mode is auto focus, means value for the picture adjusted after auto exposure.

TABLE II: AXI Signal for Write Address Channel

Signal	Width	From	Description
awid_s_inf	4	ISP.v	Write address ID. In this project, we only use this to recognize master, reordering method is not supported.
awaddr_s_inf	32	ISP.v	Write address. Address start from 32'h10000, and each pic location is 32'h10000 + 3072*pic_no.
awsize_s_inf	3	ISP.v	Burst size. Only support 3'b100 in this exercise.
awburst_s_inf	2	ISP.v	Burst type. Only INCR(2'b01) support in this Project.
awlen_s_inf	8	ISP.v	Burst length. The burst length gives the exact number of transfers in a burst.
awvalid_s_inf	1	ISP.v	Write address valid. 1 = address and control information available; 0 = address and control information not available.
awready_s_inf	1	pseudo_DRAM.v	Write address ready. 1 = slave ready; 0 = slave not ready.

TABLE III: AXI Signal for Write Data Channel

Signal	Width	From	Description
wdata_s_inf	128	ISP.v	Write data.
wlast_s_inf	1	ISP.v	Write last. This signal indicates the last transfer in a write burst.
wvalid_s_inf	1	ISP.v	Write valid. 1 = write data and strobes available; 0 = write data and strobes not available.
wready_s_inf	1	pseudo_DRAM.v	Write ready. 1 = slave ready; 0 = slave not ready.

TABLE IV: AXI Signal for Write Response Channel

Signal	Width	From	Description
bid_s_inf	4	pseudo_DRAM.v	Response ID. The BID value must match the AWID value.
bresp_s_inf	2	pseudo_DRAM.v	Write response. In this project we only issue OKAY(2'b00)
bvalid_s_inf	1	pseudo_DRAM.v	Write response valid. 1 = write response available; 0 = write response not available.
bready_s_inf	1	ISP.v	Response ready. 1 = master ready; 0 = master not ready.

TABLE V: AXI Signal for Read Address Channel

Signal	Width	From	Description
arid_s_inf	4	ISP.v	Read address ID. In this project, we only use this to recognize master, reordering method is not supported
araddr_s_inf	32	ISP.v	Read address. Each image location is started from 32'h10000 + 3072*pic_no.
arlen_s_inf	8	ISP.v	Burst length.
arsize_s_inf	3	ISP.v	Burst size. Only support 3b'100.
arburst_s_inf	2	ISP.v	Burst type. Only INCR: 2b'01 support in this Project.
arvalid_s_inf	1	ISP.v	Read address valid. 1 = address and control information valid; 0 = address and control information not valid.
arready_s_inf	1	pseudo_DRAM.v	Read address ready. 1 = slave ready; 0 = slave not ready.

TABLE VI: AXI Signal for Read Data Channel

Signal	Width	From	Description
rid_s_inf	4	pseudo_DRAM.v	Read ID tag. The RID value is generated by the slave and must match the ARID value.
rdata_s_inf	128	pseudo_DRAM.v	Read data.
rresp_s_inf	2	pseudo_DRAM.v	Read response. In this project we only issue OKAY (2'b00).
rlast_s_inf	1	pseudo_DRAM.v	Read last.
rvalid_s_inf	1	pseudo_DRAM.v	Read valid. 1 = read data available; 0 = read data not available.
rready_s_inf	1	ISP.v	Read ready. 1= master ready; 0 = master not ready.

A. Auto Focus

The auto focus algorithm primarily works by simulating different focal lengths to identify the area with the maximum focus. The image size is fixed to $3 * 32 * 32$. The specific procedure is as follows:

I. Given 3 contrasts, from 0 to 2, cut the center of image to process. The coordinate of center of image for 4 contrasts are shown bellow (count from 0):

- 0: $15 \leq x \leq 16$, $15 \leq y \leq 16$, $2 * 2$, total 4 points
- 1: $14 \leq x \leq 17$, $14 \leq y \leq 17$, $4 * 4$, total 16 points
- 2: $13 \leq x \leq 18$, $13 \leq y \leq 18$, $6 * 6$, total 36 points
- 3: ~~$12 \leq x \leq 19$, $12 \leq y \leq 19$, $8 * 8$, total 64 points~~

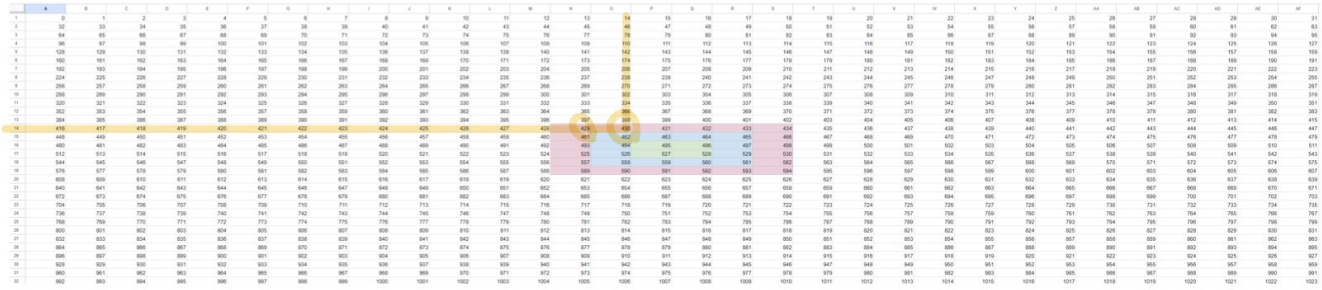


Fig. 1: The selection area for each center

II. Deal with the center area of given image for each contrasts, need to do 3 time.

$$I_{center} \in \mathbb{N}^{3\times H\times W}, \quad H = W, \quad W = \{2, 4, 6\}$$

III. Convert the selection area to grayscale. This can be represented by the following formula:

$$G_{center}(i, j) = I_{center}(i, j, 0) \cdot 0.25 + I_{center}(i, j, 1) \cdot 0.5 + I_{center}(i, j, 2) \cdot 0.25$$

The definition of each symbol:

$$[I_{center} \in \mathbb{N}^{3\times H\times W}, 0 \leq I \leq 255], \quad [G_{center} \in \mathbb{N}^{1\times H\times W}, 0 \leq G \leq 255],$$

IV. Calculate the difference between each element on the x-axis and y-axis separately, and take the absolute value. Sum up all the calculated differences.

$$D_{contrast_h*w} = \sum_{j=0}^{H-1}\sum_{i=0}^{W-2} |G_{center}(i+1, j) - G_{center}(i, j)| + \sum_{i=0}^{W-1}\sum_{j=0}^{H-2} |G_{center}(i, j+1) - G_{center}(i, j)|$$

$$D_{contrast_h*w} \in \mathbb{N}^3, h = w, w = 2, 4, 6$$

V. Divide the summed value by the processed image's area, and record this value.

$$D_{contrast_h*w} = D_{contrast_h*w}/H^2$$

VI. Repeat steps III. to V. for four contrast. Find the maximum contrast value, which is the optimal solution. If there are two or above value are same, output the smaller one's contrast value. E.g. if contrast value 0 and 2 both's solution are same but larger than 1's, please output 0.

$$max_contrast = \arg\max(D_{contrast_h*w})$$

The python and numpy-like pseudo code for color auto focus function is shown below.

B. Auto Exposure

The purpose of this algorithm is to adjust the exposure ratio of a given image, which size is fixed to 3*32*32.

$$0 \leq i, j \leq 31$$

I. First, input the entire image and the desired exposure adjustment ratio. There are 4 options, from 0 to 3, corresponding to 0.25, 0.5, 1, and 2 times the original exposure.

$$(input, ratio) = \{(0, 0.25), (1, 0.5), (2, 1), (3, 2)\}$$

II. Adjust the original image according to the chosen exposure ratio and save the adjusted picture. If any adjusted value exceeds the maximum (255), set that value to 255.

$$I_{i,j} = \begin{cases} I_{i,j} * ratio & \text{for } 0 \leq (I_{i,j} * ratio) < 256 \\ 255 & \text{other} \end{cases}$$

III. Finally, calculate the average value of the adjusted image and return this average. The detailed method is as follows:

- Multiply each element in the R and B channels by 0.25
- Multiply each element in the G channel by 0.5
- Sum all these values and divide by the original image size (1024)
- This result is the average value.

$$Avg = \frac{\sum_{i=0}^{32} \sum_{j=0}^{32} [I(i, j, 0) \cdot 0.25 + I(i, j, 1) \cdot 0.5 + I(i, j, 2) \cdot 0.25]}{32 * 32}$$

IV. Due to possible precision errors in the calculations, the answer allows a margin of error of ±1. ($|golden - ans| < 2$)

The python and numpy-like pseudo code for color auto exposure function is shown below.

C. Average of Min and Max in the Picture

In this function, you need to find out the maximum and minimum value in each RGB channels in the selected picture from the DRAM. You need to sum up three maximum values in each channel then divide this value by 3, and so does the minimum value. After you getting the average value of maximum value and minimum value, get the average value of them. The math formula for this function is shown bellow.

$$max_{R,G,B} \in 3 * N, min_{R,G,B} \in 3 * N,$$

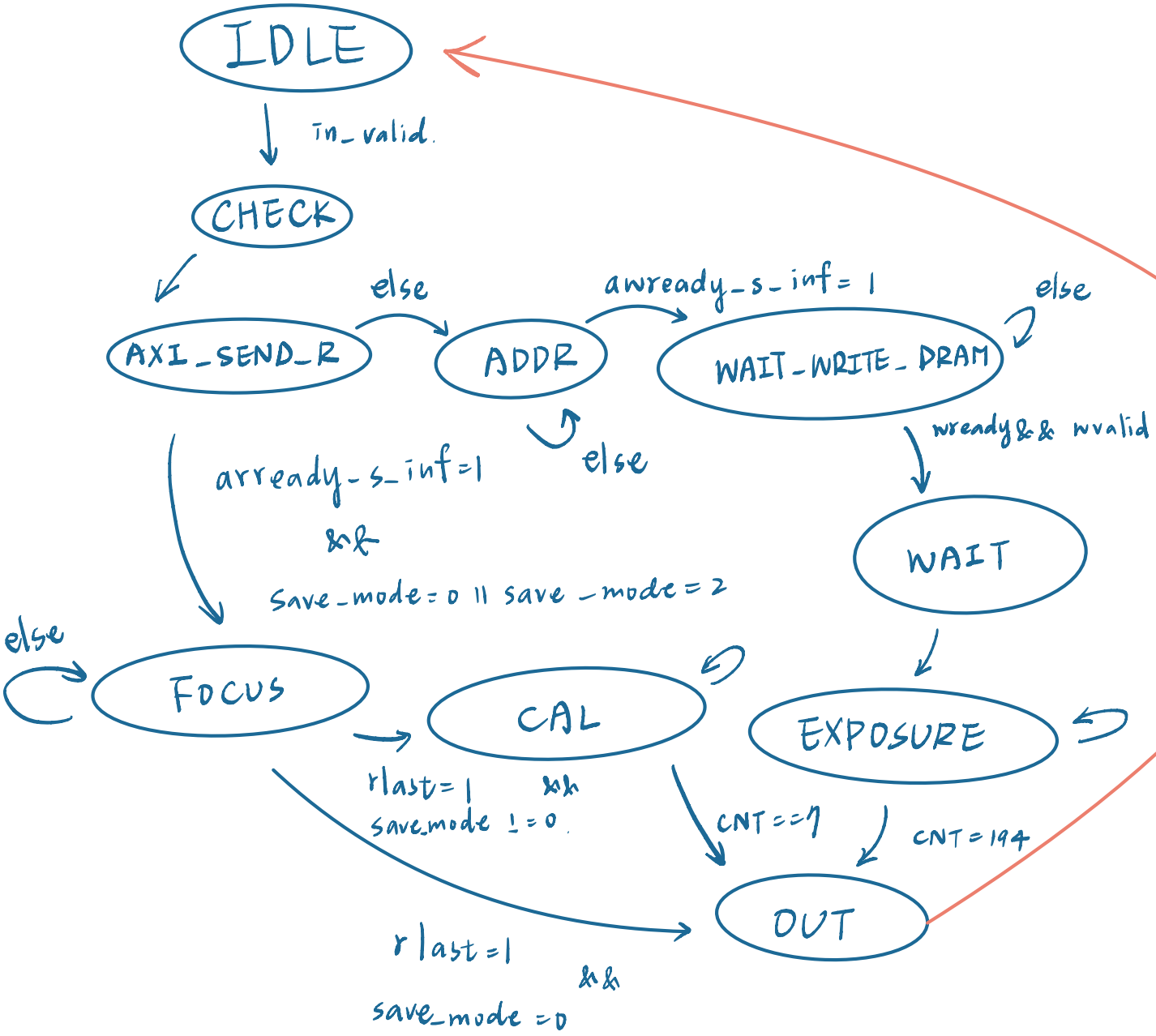
$$max_{R,G,B} = Max(Pic_{pic_idx(R,G,B)}), min_{R,G,B} = Min(Pic_{pic_idx(R,G,B)})$$

$$max = \frac{max_R + max_G + max_B}{3}, min = \frac{min_R + min_G + min_B}{3}$$

$$result = \frac{max + min}{2}$$

Here is an example for this function, and the numbers in this example are shown in decimal. We got 2, 4, 6 as the minimum value for each channel, and 80, 160, 140 for the maximum value for each channel, respectively. Now you should get the average minimum and maximum value, which is 4 and 126. Now, you can output the computation result to the PATTERN, which is 65.

FSM


IDLE
in_valid
CHECK
AXI_SEND_R
else
ADDR
awready_s_inf = 1
WAIT_WRITE_DRAM
else
else
wready && wvalid
WAIT
arready_s_inf=1
&&
save_mode = 0 || save_mode = 2
else
FOCUS
CAL
EXPOSURE
rlast=1 &&
savemode != 0
CNT==7
CNT=194
OUT
rlast=1
&&
save_mode=0